

# Lesson screens — what the portal fills in, slot by slot

Kidsverse+ — Discover (14), Spot the Mistake (16), Mission Complete (17)

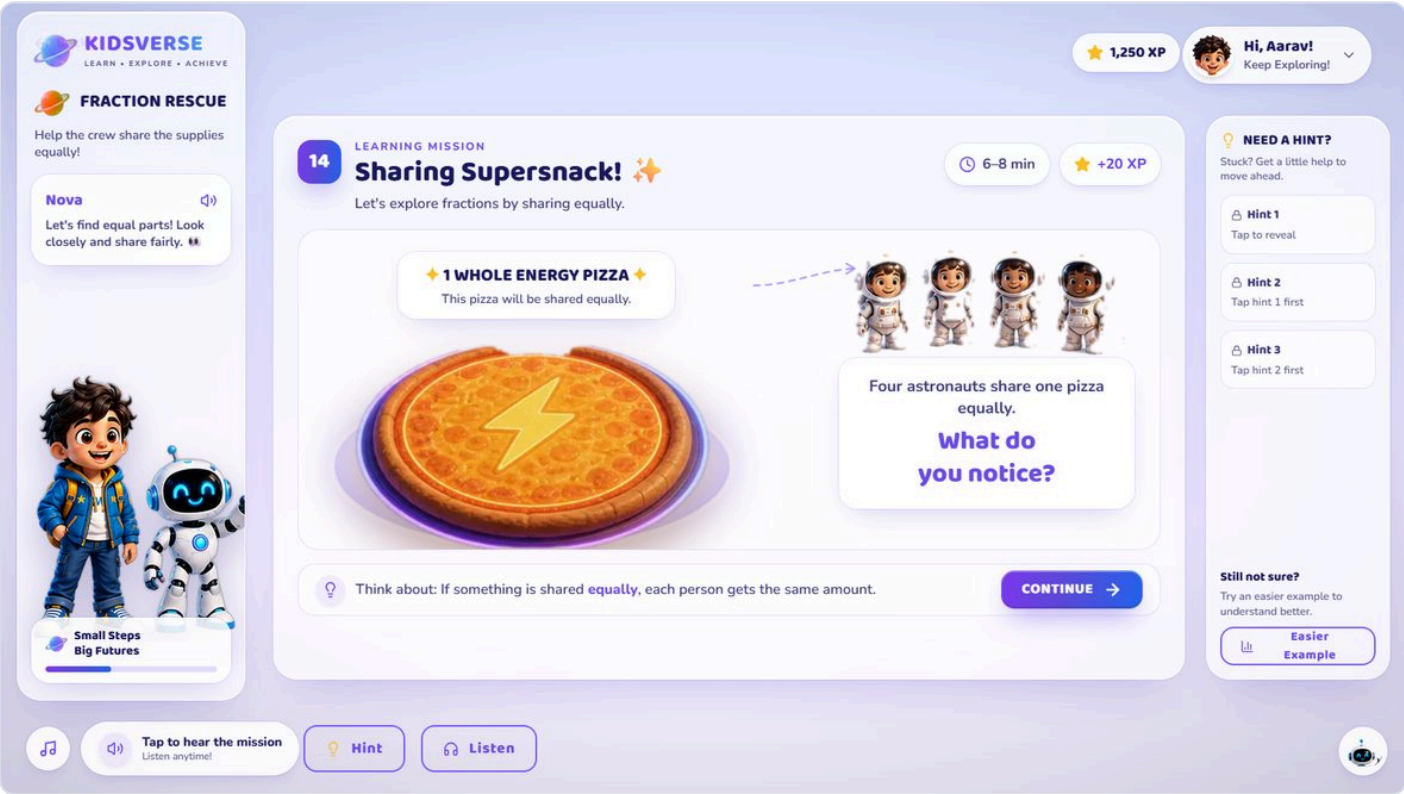
For the content-authoring portal    Date 8 September 2026    Source read from the running screens, not the designs

**The idea in one line**

Each of these screens is a **template with named holes**. The portal fills the holes; the screen draws whatever it is given. Fractions, Comprehension and Living Things all use the same three templates — only the contents of the holes change. Below is every hole, what type of thing goes in it, how big it can be, and what the screen does if it is left empty.

**Slot types:** **text** short or long text    **image** picture    **voice** Nova speaks it    **number** a number    **choice** pick from a list    **system** computed by the app, never typed by an author

## Screen 14 — Discover the teaching step

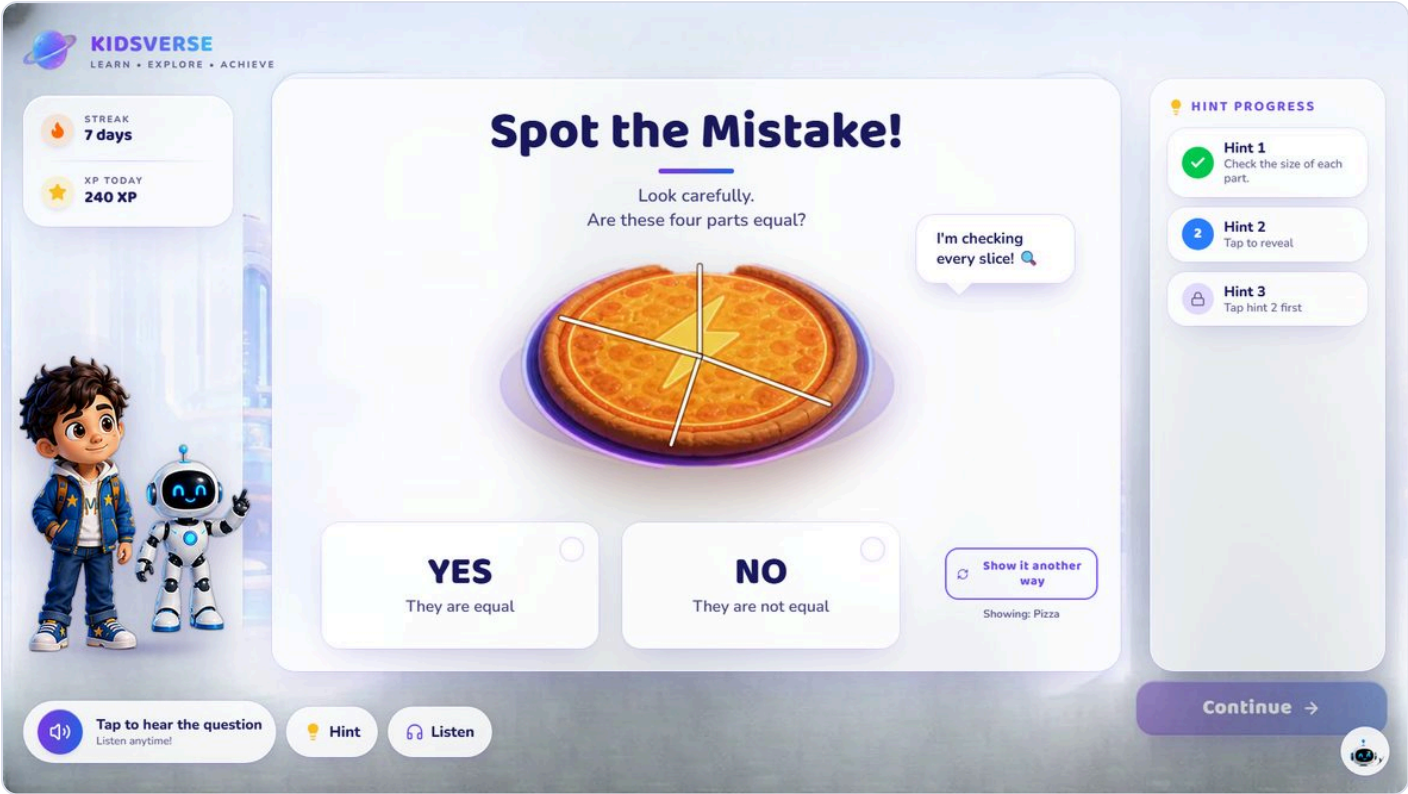


| # | Slot             | Type   | On screen right now                         | Rules                                                                      |
|---|------------------|--------|---------------------------------------------|----------------------------------------------------------------------------|
| 1 | mission.code     | text   | 14                                          | 1–2 characters. The purple badge.                                          |
| 2 | mission.title    | text   | Sharing Supersnack! ✨                       | ≤ 32 characters. One emoji allowed at the end.                             |
| 3 | mission.subtitle | text   | Let's explore fractions by sharing equally. | ≤ 80 characters, one sentence.                                             |
| 4 | mission.duration | number | 6–8 min                                     | Two numbers, minutes. Shown as a range.                                    |
| 5 | mission.xp       | number | +20 XP                                      | Whole number. The same number is paid on screen 17 — author it once, here. |
| 6 | rail.title       | text   | FRACTION RESCUE                             | ≤ 24 characters. Rendered in caps.                                         |

|    |                      |        |                                                                   |                                                                                                                          |
|----|----------------------|--------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 7  | rail.blurb           | text   | Help the crew share the supplies equally!                         | ≤ 60 characters.                                                                                                         |
| 8  | model.title          | text   | 1 WHOLE ENERGY PIZZA                                              | ≤ 28 characters. Rendered in caps.                                                                                       |
| 9  | model.caption        | text   | This pizza will be shared equally.                                | ≤ 60 characters.                                                                                                         |
| 10 | model.image          | image  | the pizza                                                         | The main teaching picture. 560×340 design px, PNG with transparency, ship at 2× (1120×680).                              |
| 11 | crew.images          | image  | four astronauts                                                   | A set. Count must equal the number in the prompt (4 here, 2 in the easier version). Each 130×150 design px, transparent. |
| 12 | prompt.statement     | text   | Four astronauts share one pizza equally.                          | ≤ 80 characters. Plain fact.                                                                                             |
| 13 | prompt.question      | text   | What do you notice?                                               | ≤ 40 characters. Rendered large.                                                                                         |
| 14 | think_about          | text   | If something is shared equally, each person gets the same amount. | ≤ 120 characters. One bolded key word allowed ( <b>equally</b> ).                                                        |
| 15 | hints[1..3]          | text   | 3 hints, opened in order                                          | Exactly three, each ≤ 100 characters. The child can only open hint 2 after hint 1.                                       |
| 16 | easier.enabled       | choice | yes — two astronauts                                              | yes/no. If yes, the portal also supplies <code>easier.crew_count</code> and <code>easier.statement</code> .              |
| 17 | nova.speech          | text   | Let's find equal parts! Look closely and share fairly. 🗣️         | ≤ 90 characters. The speech bubble beside Nova.                                                                          |
| 18 | nova.voice           | voice  | "Tap to hear the mission"                                         | What Nova reads aloud. Script text ≤ 400 characters; the app speaks it. Optional MP3 to override the voice.              |
| 19 | Continue → next step | system | goes to screen 16                                                 | Not authored. The mission's step order decides it.                                                                       |

Not authored on this screen: the child's name, XP total, streak, the "LEARNING MISSION" eyebrow, the button labels. Those are fixed or come from the child's account.

Screen 16 — Spot the Mistake the check-for-understanding step



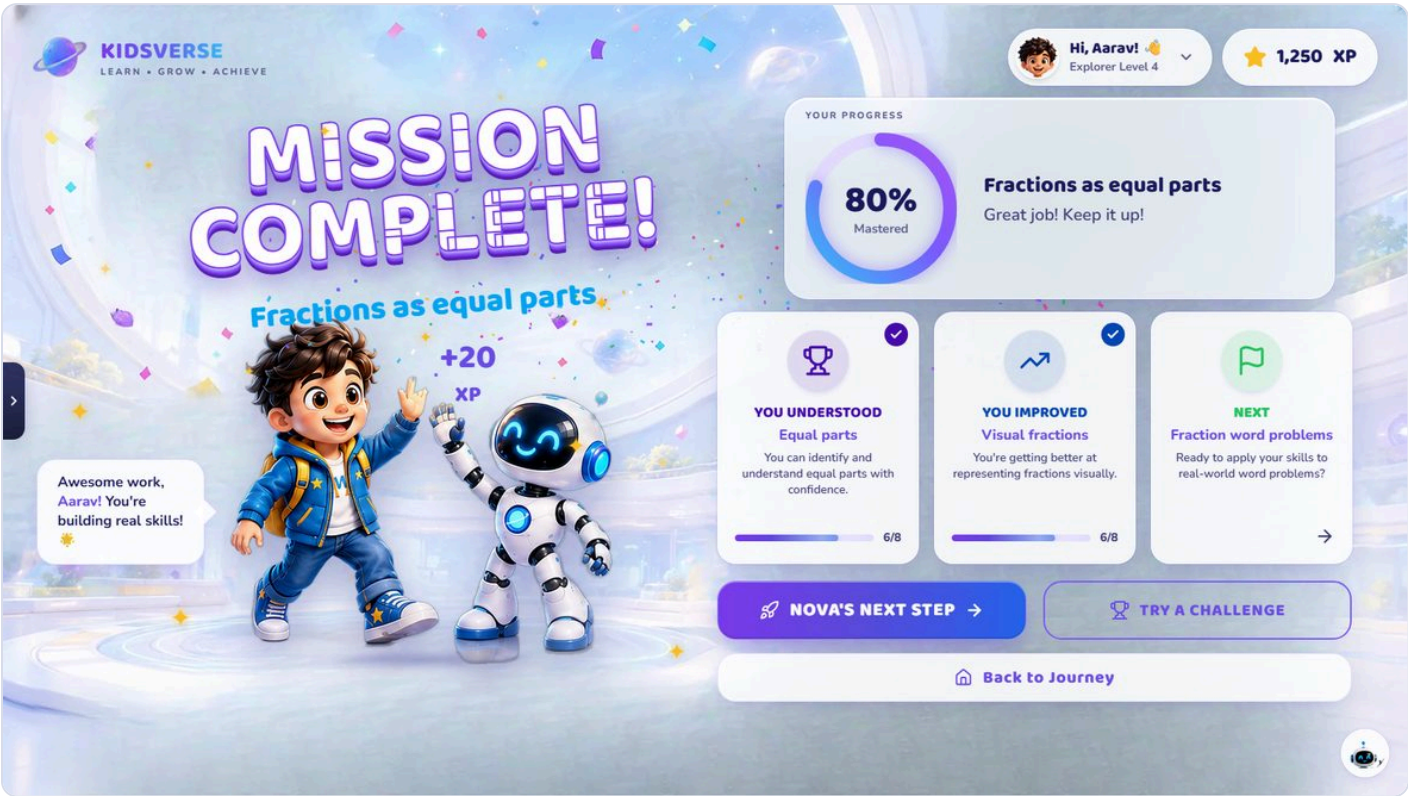
The one thing that is different about this screen

It carries **three pictures of the same idea**, not one. “Show it another way” swaps the pizza for a chocolate bar, then a number line. The question, the options and the right answer never change; only the picture does — and **each picture has its own three hints**, because “look at the crust” means nothing on a number line. So the portal has to collect hints per picture, not per screen.

| # | Slot                     | Type   | On screen right now                                                                 | Rules                                                                                   |
|---|--------------------------|--------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1 | title                    | text   | Spot the Mistake!                                                                   | ≤ 24 characters.                                                                        |
| 2 | instruction              | text   | Look carefully. Are these four parts equal?                                         | ≤ 80 characters. Two short sentences work best.                                         |
| 3 | models[]                 | choice | Pizza, Chocolate bar, Number line                                                   | 1 to 3 alternatives. Each has the four fields below. The first is shown by default.     |
| 4 | models[n].label          | text   | Pizza                                                                               | ≤ 16 characters. Shown as “Showing: Pizza”.                                             |
| 5 | models[n].image          | image  | the unevenly-cut pizza                                                              | The picture <i>with the mistake in it</i> . 520×340 design px, transparent, ship at 2×. |
| 6 | models[n].hints[1..3]    | text   | Check the size of each part · Look at the crust lengths · Compare the toppings area | Exactly three per picture, each ≤ 80 characters, in the order they unlock.              |
| 7 | models[n].feedback.wrong | text   | Hmm, look again at the crust. Is every slice the same size?                         | ≤ 100 characters. Should name something in <i>this</i> picture.                         |

|    |                     |        |                                                        |                                                                                                             |
|----|---------------------|--------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 8  | options[]           | choice | YES / NO                                               | 2 to 4 options. Each has a big label ( $\leq 8$ chars) and a sub-line ( $\leq 30$ chars): "They are equal". |
| 9  | answer              | choice | NO                                                     | Which option is correct. One only.                                                                          |
| 10 | feedback.correct    | text   | Great observation! Not all four slices are the same. ✦ | $\leq 100$ characters. Shown once the right option is picked.                                               |
| 11 | nova.speech         | text   | I'm checking every slice! 🔍                            | $\leq 60$ characters.                                                                                       |
| 12 | nova.voice          | voice  | "Tap to hear the question"                             | Script $\leq 300$ characters. Usually the instruction read aloud.                                           |
| 13 | xp_on_correct       | number | +10 XP                                                 | Paid on the first correct answer only. Whole number.                                                        |
| 14 | Hint progress panel | system | Hint 1 ✓ · Hint 2 · Hint 3 locked                      | Not authored. Drawn from the hints above and from what the child has opened.                                |

Screen 17 — Mission Complete the outcome



Most of this screen is not content

The XP, the mastery ring, the 6/8 progress bars and the ticks are all **about the child**, so they must come from the child's record, never from the portal. Today the ring and the bars are fixed numbers in the code — the same 80% for every child — which is exactly the thing this split fixes. The portal owns the words; the app owns the numbers.

| # | Slot                    | Type   | On screen right now                                          | Rules                                                                                                |
|---|-------------------------|--------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1 | topic.label             | text   | Fractions as equal parts                                     | ≤ 40 characters. Appears twice: under the headline and in the progress card.                         |
| 2 | nova.speech             | text   | Awesome work, {name}! You're building real skills! 🌟         | ≤ 90 characters. {name} is replaced with the child's name.                                           |
| 3 | progress.encouragement  | text   | Great job! Keep it up!                                       | ≤ 40 characters.                                                                                     |
| 4 | outcomes[ ]             | choice | 3 cards                                                      | Up to three. Each has the four fields below. Order left to right.                                    |
| 5 | outcomes[n].kind        | choice | understood / improved / next                                 | One of the three. Sets the icon and colour; the first two get a tick, "next" gets an arrow.          |
| 6 | outcomes[n].heading     | text   | YOU UNDERSTOOD                                               | ≤ 18 characters, caps.                                                                               |
| 7 | outcomes[n].skill       | text   | Equal parts                                                  | ≤ 28 characters. Should match a skill the portal has defined, so the bar underneath can be computed. |
| 8 | outcomes[n].description | text   | You can identify and understand equal parts with confidence. | ≤ 110 characters.                                                                                    |

|    |              |        |                                    |                                                                                           |
|----|--------------|--------|------------------------------------|-------------------------------------------------------------------------------------------|
| 9  | next_step    | choice | the fraction word-problems mission | Which mission "Nova's Next Step" opens. Pick from existing missions.                      |
| 10 | XP awarded   | system | +20 XP                             | Comes from mission.xp on screen 14. Never typed here.                                     |
| 11 | Mastery ring | system | 80% Mastered                       | Computed from the child's results. Today a literal; will be real once results are stored. |
| 12 | Skill bars   | system | 6/8, 6/8                           | Computed per skill from the child's results. Same note.                                   |



## What one step looks like on the wire

One JSON object per screen. This is the actual Fractions content, in the shape the portal would send and the screen would read. Every key above appears here by the same name.

```
{
  "mission": { "code": "14", "title": "Sharing Supersnack!", "emoji": "🍪",
    "subtitle": "Let's explore fractions by sharing equally.",
    "duration_min": [6, 8], "xp": 20,
    "rail": { "title": "Fraction Rescue", "blurb": "Help the crew share the supplies equally!" } },

  "discover": {
    "model": { "title": "1 Whole Energy Pizza", "caption": "This pizza will be shared equally.",
      "image": "img/fractions/pizza-whole.png" },
    "crew": { "count": 4, "images": ["img/crew/astro-1.png", "..."] },
    "prompt": { "statement": "Four astronauts share one pizza equally.", "question": "What do you notice?" },
    "think_about": "If something is shared *equally*, each person gets the same amount.",
    "hints": [ "Look at the pizza. How many equal parts could we make?",
      "If we cut it into 4 equal parts, what fraction does each astronaut get?",
      "Each astronaut gets one part out of four equal parts." ],
    "easier": { "enabled": true, "crew_count": 2, "statement": "Two astronauts share one pizza equally." },
    "nova": { "speech": "Let's find equal parts! Look closely and share fairly.",
      "voice": "Let's find equal parts. Look closely at the pizza and share it fairly." }
  },

  "check": {
    "title": "Spot the Mistake!", "instruction": "Look carefully. Are these four parts equal?",
    "models": [
      { "key": "pizza", "label": "Pizza", "image": "img/fractions/pizza-uneven.png",
        "hints": ["Check the size of each part.", "Look at the crust lengths.", "Compare the toppings area."],
        "feedback_wrong": "Hmm, look again at the crust. Is every slice the same size?" },
      { "key": "bar", "label": "Chocolate bar", "image": "...", "hints": [...], "feedback_wrong": "..." },
      { "key": "line", "label": "Number line", "image": "...", "hints": [...], "feedback_wrong": "..." }
    ],
    "options": [ { "key": "yes", "label": "YES", "sub": "They are equal" },
      { "key": "no", "label": "NO", "sub": "They are not equal" } ],
    "answer": "no",
    "feedback_correct": "Great observation! Not all four slices are the same.",
    "xp_on_correct": 10,
    "nova": { "speech": "I'm checking every slice!", "voice": "Look carefully. Are these four parts equal?" }
  },

  "complete": {
    "topic_label": "Fractions as equal parts", "encouragement": "Great job! Keep it up!",
    "nova": { "speech": "Awesome work, {name}! You're building real skills!" },
    "outcomes": [
      { "kind": "understood", "heading": "You understood", "skill": "Equal parts",
        "description": "You can identify and understand equal parts with confidence." },
      { "kind": "improved", "heading": "You improved", "skill": "Visual fractions",
        "description": "You're getting better at representing fractions visually." },
      { "kind": "next", "heading": "Next", "skill": "Fraction word problems",
        "description": "Ready to apply your skills to real-world word problems?" }
    ],
    "next_step": "mission:fractions-word-problems"
  }
}
```

Underlined values are uploaded files; the portal stores the file and puts its path here. Green values are voice scripts — text the app speaks. Everything else is typed into a form field.

## Images

| Slot                           | Size (design px)                                                                                                                                                                                          | Ship at    | Notes                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|
| Discover · <u>model.image</u>  | 560 × 340                                                                                                                                                                                                 | 1120 × 680 | PNG, transparent background. The hero picture of the step.                        |
| Discover · <u>crew.images</u>  | 130 × 150 each                                                                                                                                                                                            | 260 × 300  | One file per character. Count is set by the prompt.                               |
| Check · <u>models[n].image</u> | 520 × 340                                                                                                                                                                                                 | 1040 × 680 | PNG, transparent. This is the <i>flawed</i> picture; the mistake must be visible. |
| All                            | The canvas is 1672 × 941. Everything ships at 2× for sharp tablets. Max 600 KB per file. No baked-in text — captions are typed into the slot beside the picture so they can be translated and read aloud. |            |                                                                                   |

### Why “three pictures of one idea” matters for the portal

The app already knows how to draw a pizza, a chocolate bar and a number line for any fraction from a single set of numbers — it did that so that swapping the picture can never change the answer. So for fraction steps the portal can ask for **the split** (four parts, one is bigger) rather than three drawings, and the app draws all three. For other subjects the portal collects three uploaded pictures. Both paths land in the same `models[]` slot.

## Voice

- Every `nova.voice` is a **script**, typed as text. The app speaks it with the same voice on every screen, so the author never records anything.
- Write it as it should be *said*, not as it is written: numbers as words, no emoji, no caps.
- An optional MP3 can be attached to any voice slot to override the generated speech — for a character moment, or a language the voice does not handle.
- Where it plays: Discover → “Tap to hear the mission”; Check → “Tap to hear the question”; the Listen button on both reads the on-screen text.



## How this maps onto the authoring journey

The proposed portal sequence — Topic → Learning Outcome → Teaching Method → Alternate Method → Re-check → Practice → Mastery Check → Complete — lands on these three screens like this:

| Portal stage              | Screen                                               | What it fills                                                                                                              |
|---------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Learning Outcome          | 17 · Mission Complete                                | <code>topic.label1</code> , the three <code>outcomes[ ]</code> . Written first because everything else exists to reach it. |
| Teaching Method           | 14 · Discover                                        | The model, the crew, the prompt, think-about, hints, Nova.                                                                 |
| Alternate Teaching Method | 14 · Discover (easier) and 16 · models[2], models[3] | The easier version of the same step, and the bar and number-line pictures with their own hints.                            |
| Re-check                  | 16 · Spot the Mistake                                | The flawed picture, the options, the answer, the feedback.                                                                 |
| Practice                  | 16, repeated                                         | More check steps of the same shape. The template does not change; the portal adds steps.                                   |
| Mastery Check             | Test Arena (19–21)                                   | Outside these three screens. Same slot pattern: a question, options, an answer.                                            |
| Complete                  | 17                                                   | Already filled at the Learning Outcome stage; the app adds the numbers.                                                    |

## What the author sees, in order

1. **Mission card** — code, title, subtitle, duration, XP, rail title and blurb. Seven fields.
2. **Outcome** — topic label, up to three outcome cards, which mission comes next.
3. **Teach it** — model title and caption, upload the picture, crew count and pictures, the statement and question, think-about line, three hints, Nova’s bubble and script. Then the easier version: on/off, crew count, statement.
4. **Check it** — title and instruction, then *one tab per picture* (Pizza / Bar / Number line): upload, three hints, wrong-answer feedback. Then the options, the right one, the correct-answer feedback, Nova’s bubble and script.
5. **Preview** — the three screens rendered with exactly this content, on the real templates, before publish.

### Two rules for the portal that save trouble later

**Every text slot has a hard character limit**, listed above. The screens are fixed-layout for a tablet; a title that is one word too long wraps onto the picture. Enforce the limits in the form, not in review.

**Nothing about the child goes in the portal.** No XP totals, no mastery, no progress bars, no names. If a slot describes the learner rather than the lesson, it is the app’s job.